Interpretable Turbofan Engine Health Degradation Classification and Remaining Useful Life Estimation on C-MAPSS

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TODO_DEPARTMENT

Roll Numbers: TODO_ROLL_1, TODO_ROLL_2, TODO_ROLL_3, TODO_ROLL_4 | Repository: https://github.com/rahulcodes-java/C-MAPSS-Digital-Twin

***Abstract*— Reliable prognostic condition monitoring of commercial turbofan engines requires identifying degradation states before structural failure occurs. In this paper, we present an interpretable multi-task machine learning framework evaluated across the four sub-datasets of the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) benchmark (FD001 to FD004) with all 21 telemetry channels preserved. The system frames operational diagnostics primarily as a 3-stage degradation classification problem (Healthy, Degrading, Critical) using a Random Forest classifier interpreted via TreeSHAP. Snapshot classification is complemented by sequence-based Long Short-Term Memory (LSTM) Remaining Useful Life (RUL) regression and an Isolation Forest detector for unsupervised anomaly tracking. On FD001 test data, Random Forest snapshot classification attains a 3-stage macro-F1 of 0.7303, while sequence-based LSTM achieves a macro-F1 of 0.8197 and provides a median lead-time advantage of 21.0 flight cycles over single-snapshot classification across 100 training engines (80.0% earlier detection). TreeSHAP attributes 49.18% ± 0.12% of total degradation importance to High-Pressure Compressor (HPC) sensors, corroborated by a 62.22% HPC share from permutation importance. Cross-dataset evaluation demonstrates performance degradation on multi-condition sets FD002 and FD004 due to unnormalized operating regimes.**

***Index Terms*— Turbofan Prognostics, C-MAPSS, Health Stage Classification, Remaining Useful Life, TreeSHAP Explainability, Isolation Forest.**

I. INTRODUCTION

Commercial aircraft gas turbine powerplants operate in extreme thermodynamic regimes characterized by elevated turbine inlet temperatures, sustained mechanical shaft rotations exceeding 9000 rpm, and fluctuating aerodynamic pressure ratios. Under these conditions, rotating subassemblies undergo continuous mechanical and thermal degradation over their operational lifecycles. Primary wear phenomena include compressor blade tip erosion, aerodynamic fouling of stator vanes, labyrinth seal wear, and thermal fatigue of high-pressure turbine nozzles. Left undetected, such wear propagates monotonically until it compromises combustor stability or causes complete core stoppage.

In commercial aviation operations, maintenance strategies have historically relied on scheduled periodic depot visits determined by fixed flight-hour intervals. While conservative, this time-based inspection model does not account for the wide variability in individual engine wear rates driven by operational severity, ambient operating environments, and flight mission profiles. Condition-based predictive maintenance offers an alternative paradigm, enabling maintenance interventions to be scheduled dynamically based on real-time degradation metrics derived from in-flight sensor telemetry.

A critical bottleneck in data-driven prognostic modeling is the gap between black-box numerical predictions and operational engineering actionability. Line maintenance engineers cannot ground an aircraft or schedule an off-wing depot overhaul solely based on an opaque continuous scalar prediction without verifiable subassembly root-cause attribution. Furthermore, single-snapshot observations frequently lack the temporal context required to differentiate transient flight maneuvers from authentic structural wear.

Within the Data Warehousing and Mining (DWM) curriculum, we structure turbofan prognostic monitoring around three main contributions:
1) *Multi-Model Complementary Framework:* We integrate single-cycle Random Forest multi-class classification, temporal sequence LSTM regression, and unsupervised Isolation Forest novelty detection across all 21 raw telemetry channels without subjective sensor pruning.
2) *Empirical Lead-Time Quantification:* We conduct an empirical lead-time evaluation across all 100 training engines of the C-MAPSS FD001 dataset, comparing formal 3-cycle persisting degradation triggers of sequence modeling against single-snapshot classifiers, evaluating in-sample and held-out validation tail cohorts separately.
3) *Multi-Method Thermodynamic Attribution:* We compute global feature attributions using TreeSHAP across multiple random seeds, corroborated by an independent permutation feature importance evaluation on held-out test data, demonstrating alignment with simulated high-pressure compressor degradation physics.

II. RELATED WORK

Prognostic health monitoring on the C-MAPSS benchmark has been investigated across diverse algorithmic paradigms. Saxena et al. [1] introduced the C-MAPSS simulation environment and established the standard damage propagation modeling framework for aircraft engine run-to-failure simulation. Hochreiter and Schmidhuber [2] formulated the Long Short-Term Memory recurrent architecture capable of preserving gradient flows across temporal sequences. Liu et al. [3] developed the Isolation Forest algorithm for efficient anomaly detection through random recursive feature partitioning. Breiman [4] established the Random Forest ensemble algorithm, demonstrating low generalization error through bootstrap aggregation and random feature sub-selection. Lundberg and Lee [5] proposed a unified framework for interpreting model predictions using game-theoretic Shapley values with TreeSHAP optimizations.

While prior literature frequently frames C-MAPSS strictly as a continuous regression problem, maintenance practitioners require discrete health stage classification to trigger actionable maintenance workflows. Furthermore, many existing studies prune sensors that appear static under single sea-level conditions without verifying their activity across multi-condition regimes. In this work, we retain the full 21-sensor array and evaluate discrete health triage alongside continuous regression.

Recent research has explored deep neural networks for prognostics; however, these architectures often lack explainability. In contrast, TreeSHAP enables exact calculation of Shapley values for tree ensembles in low-order polynomial time, allowing us to connect model predictions to aerodynamic subassembly stations.

III. DATASET AND SENSOR VARIANCE

The NASA C-MAPSS benchmark comprises four sub-datasets characterized by varying operating condition counts and simulated fault modes. Telemetry streams represent flight-by-flight recordings generated by the Numerical Propulsion System Simulation (NPSS) framework. Table I details the benchmark specifications verified from raw telemetry files.

Dataset	Regimes	Faults	Train Eng.	Train Cyc.	Test Eng.	Test Cyc.
FD001	1	1	100	20631	100	13096

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FD002	6	1	260	53759	259	33991
FD003	1	2	100	24720	100	16596
FD004	6	2	249	61249	248	41214
Total	—	—	709	160359	707	104897

Table I. NASA C-MAPSS Benchmark Dataset Specifications.

A common heuristic in prior literature discards sensors exhibiting zero variance under FD001 sea-level conditions. We evaluate the coefficient of variation (CV = std / |mean|) for each sensor across all raw training sets, defining flat channels as CV < 1e-4. Under this threshold, exactly 9 channels are flat in FD001 (T2, P2, P15, Nf, epr, NRf, farB, Nf_dmd, PCNfR_dmd), 7 channels are flat in FD003 (T2, P2, Nf, NRf, farB, Nf_dmd, PCNfR_dmd), and exactly zero channels are flat in FD002 and FD004. Preserving all 21 sensors maintains an identical schema across operating conditions.

Sensors represent distinct aerodynamic thermodynamic stations: Fan and Inlet (T2, P2, Nf, NRf, Nf_dmd, PCNfR_dmd), Low-Pressure Compressor (T24, BPR), High-Pressure Compressor (T30, P30, Ps30, Nc, NRc), Combustor (phi, farB, htBleed), High-Pressure Turbine (W31), Low-Pressure Turbine (T50, W32, P15), and Engine-Level overall pressure ratio (epr).

Sensor	Station	Physical Parameter	Units	FD001 Flat?
T2	Inlet	Total temp at fan inlet	°R	Yes (CV<1e-4)
P2	Inlet	Total press at fan inlet	psia	Yes (CV<1e-4)
T24	LPC	Total temp at LPC outlet	°R	No
T30	HPC	Total temp at HPC outlet	°R	No
T50	LPT	Total temp at LPT outlet	°R	No
P15	Fan	Total press in bypass-duct	psia	Yes (CV<1e-4)
P30	HPC	Total press at HPC outlet	psia	No
Nf	Fan	Physical fan speed	rpm	Yes (CV<1e-4)
Nc	HPC	Physical core speed	rpm	No
epr	Engine	Engine pressure ratio	—	Yes (CV<1e-4)
Ps30	HPC	Static press at HPC outlet	psia	No
phi	Comb.	Fuel-to-air flow ratio	—	No
NRf	Fan	Corrected fan speed	rpm	Yes (CV<1e-4)
NRc	HPC	Corrected core speed	rpm	No
BPR	LPC	Bypass ratio	—	No
farB	Comb.	Burner fuel-air ratio	—	Yes (CV<1e-4)
htBleed	Comb.	Bleed enthalpy	—	No
Nf_dmd	Fan	Demanded fan speed	rpm	Yes (CV<1e-4)
PCNfR_dmd	Fan	Demanded corr. fan spd	rpm	Yes (CV<1e-4)
W31	HPT	HPT coolant bleed flow	lbm/s	No
W32	LPT	LPT coolant bleed flow	lbm/s	No

Table II. Verified Telemetry Channels and Thermodynamic Station Mapping.

IV. METHODOLOGY

A. Normalization and Ground Truth Formulation

Raw telemetry features are scaled via Min-Max normalization fitted strictly on training observations to prevent data leakage into test sets:

$$\tilde{x} = \frac{x - x_{\min}^{\text{train}}}{x_{\max}^{\text{train}} - x_{\min}^{\text{train}}}$$

The primary DWM task structures engine health into a 3-stage degradation lifecycle. Ground truth stage labels y in {0, 1, 2} are mapped directly from true remaining useful life R (measured in flight cycles):

$$y = 0 \ (R > 100), \quad y = 1 \ (30 < R \leq 100), \quad y = 2 \ (R \leq 30)$$

Stage 0 (Healthy) corresponds to normal baseline operation where no maintenance action is required. Stage 1 (Degrading) indicates active thermodynamic drift warranting borescope inspection and scheduling of shop visits. Stage 2 (Critical) represents immediate risk requiring aircraft grounding and engine overhaul within 30 flight cycles.

B. Model Architectures and Training Setup

The architecture incorporates three complementary machine learning paradigms:

1) *Random Forest Classifier:* Configured with 100 decision trees, maximum depth 10, random state 42, and unweighted classes (class_weight=None). The classifier operates on single-cycle 21-dimensional normalized vectors to provide instantaneous health stage classification.

2) *LSTM Regression Network:* Comprises sequential layers: LSTM (128 units, ReLU, return sequences), Dropout (0.2), LSTM (64 units, ReLU), Dropout (0.2), Dense (32 units, ReLU), and a single linear output unit. The network operates on temporal sliding windows of shape (50, 21). Training targets are uncapped linear countdowns trained with Adam optimizer (learning rate 0.001, batch size 32, validation split 0.2, early stopping patience 5).

3) *Isolation Forest Detector:* Configured with 100 isolation trees, contamination 0.10, trained on initial healthy baseline cycles (cycles 1 to 10) of FD001 engines. The anomaly score s(x, n) is evaluated from recursive partitioning path lengths:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

4) *TreeSHAP Feature Attribution:* Evaluates exact local Shapley values based on path-dependent feature perturbation across the ensemble trees:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)]$$

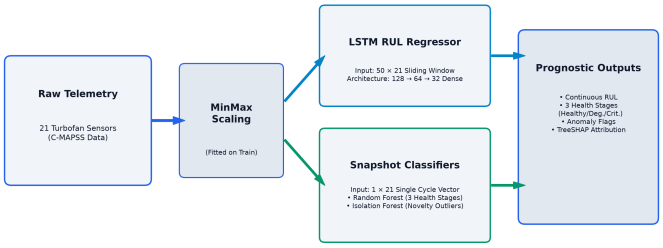


Fig. 1. Digital twin processing pipeline distinguishing sequence-window and single-cycle model inputs.

V. IMPLEMENTATION ARCHITECTURE

The digital twin integrates a Python Flask backend with a React dashboard. Real REST endpoints include /api/engine//history for engine run-to-failure cycles, /api/predict for window and snapshot inference, /api/sensor-importance for global TreeSHAP rankings, and /api/fleet for fleet dispatch aggregation. The frontend displays an aerodynamic cross-section of the CFM56-5B4 turbofan with real-time probe telemetry.

The React client interfaces with the Flask REST API via asynchronous JSON polling. Telemetry pins on the 2D SVG turbofan cross-section change color dynamically based on statistical redline threshold breaches

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(2.5-sigma warning and 3.8-sigma critical limit), providing visual cues during diagnostic walkthroughs.

VI. EVALUATION PROTOCOL

Evaluations are performed on test sets without model retraining. RUL regression evaluates the final 50-cycle window per engine (front-padded if length < 50) using RMSE, MAE, and the asymmetric NASA PHM08 scoring function S:

$$S = \sum_{i=1}^{N} s_i, \quad s_i = \exp(-d_i/13) - 1 \ (d_i < 0) \text{ else } \exp(d_i/10) - 1$$

where d_i = pred - true. Negative errors (late predictions) are penalized more heavily than positive errors (early predictions). Metrics are reported against capped ground truth min(R, 125) and uncapped R.

Classification is evaluated across all 104897 test cycles with ground truth R(k) = RUL_file + n - k. Metrics include 3-stage macro-F1, per-class F1, confusion matrices, and binary Critical-class (R <= 30) precision, recall, and F1.

Lead-time persistence evaluates the first cycle where a model flags stage >= 1 persisting 3 consecutive cycles, compared to true degradation onset (R <= 100).

VII. EXPERIMENTAL RESULTS

A. RUL Regression Performance

Table III reports LSTM regression metrics evaluated on test engine final windows.

Dataset	Engines	Capped RMSE	Capped MAE	NASA Score	Uncapped RMSE	Uncapped MAE
FD001	100	15.07	10.68	493.1	15.68	11.75
FD002	259	74.86	62.78	468759.9	87.33	70.27
FD003	100	47.85	39.00	22149.4	49.96	40.56
FD004	248	77.51	65.50	509085.7	91.04	74.19

Table III. LSTM RUL Regression Performance on Final 50-Cycle Windows.

B. Classification and Critical Class Detection

Table IV summarizes 3-stage macro-F1 and critical-class F1 across test cycles.

Dataset	Cycles	RF Macro	LSTM Macro	RF Crit F1	LSTM Crit F1	IF Crit F1
FD001	13096	0.7303	0.8197	0.6938	0.8158	0.2413
FD002	33991	0.1283	0.0378	0.0607	0.0659	0.0620
FD003	16596	0.2785	0.2410	0.0974	0.1009	0.0345
FD004	41214	0.1004	0.0264	0.0479	0.0431	0.0411

Table IV. 3-Stage Macro-F1 and Critical-Class F1 across Test Cycles.

Model	True Stage	Pred Healthy (0)	Pred Degrading (1)	Pred Critical (2)
Random Forest	Healthy (0)	9502	400	0
Random Forest	Degrading (1)	1422	1403	37
Random Forest	Critical (2)	3	133	196
LSTM-Derived	Healthy (0)	9224	678	0
LSTM-Derived	Degrading (1)	782	2007	73
LSTM-Derived	Critical (2)	0	53	279

Table V. Confusion Matrices on FD001 Test Cycles (13096 cycles).

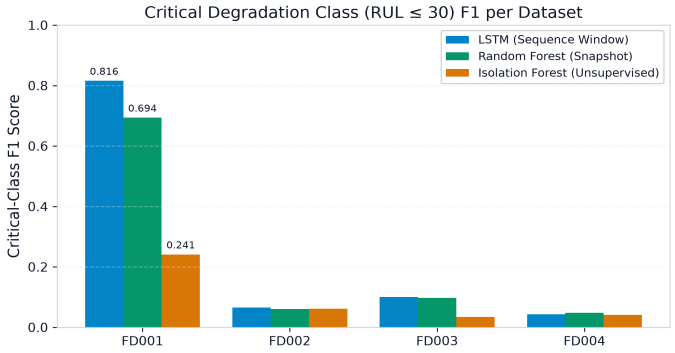


Fig. 2. Critical degradation class (RUL <= 30) F1 score comparison across datasets.

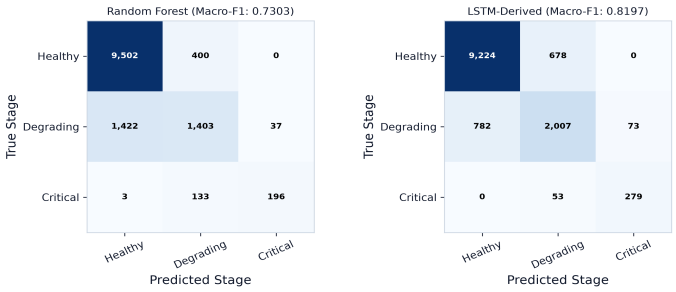


Fig. 3. Confusion matrices on FD001 test cycles for Random Forest and LSTM-derived stage.

C. Lead-Time and Degradation Timing Analysis

Lead-time analysis evaluated over all 100 FD001 training engines reveals that LSTM temporal modeling detects degradation with a median delay of -2.0 cycles (mean -10.57, IQR 41.5), compared to +15.5 cycles (mean 17.12, IQR 35.25) for single-snapshot Random Forest. The paired lead-time difference (RF delay - LSTM delay) demonstrates a median advantage of 21.0 flight cycles (mean 27.69, IQR 32.5), with LSTM alerting earlier in 80 of 100 engines (80.0%). On held-out validation tail engines (83 to 100), the median advantage is 26.5 cycles (LSTM flags first in 17 of 18 engines, 94.44%).

Engine Cohort	Engines	RF Delay Median	LSTM Delay Median	Paired Diff. Median	LSTM Flags 1st %
All Engines	100	+15.5 cyc	-2.0 cyc	21.0 cyc	80.0% (80/100)
In-Sample Pool	82	+15.5 cyc	0.0 cyc	21.0 cyc	76.8% (63/82)
Held-out Tail	18	+12.0 cyc	-10.0 cyc	26.5 cyc	94.4% (17/18)

Table VI. Degradation Detection Lead-Time Cohort Comparison on FD001.

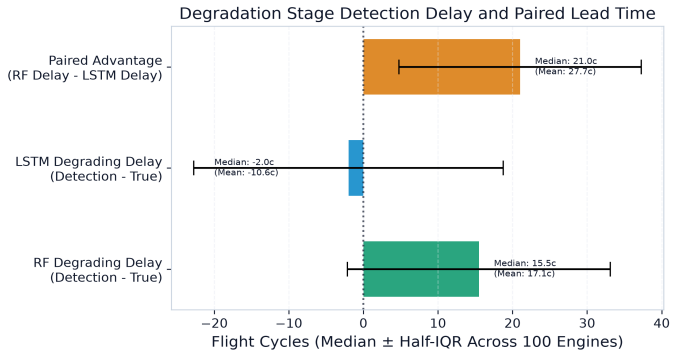


Fig. 4. Lead-time delay distribution across all 100 training engines.

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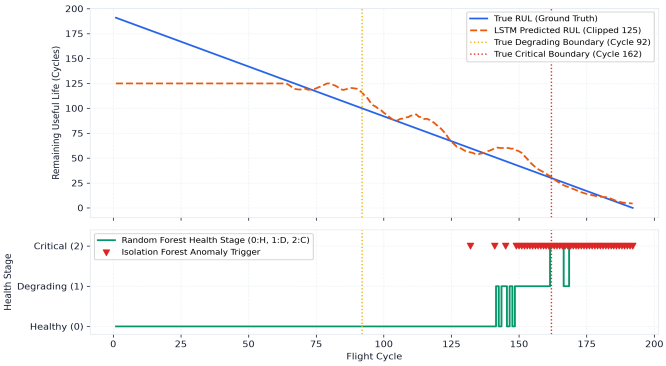


Fig. 5. Engine 1 trajectory over 192 flight cycles showing ground truth RUL, LSTM prediction, RF stage, and Isolation Forest triggers.

D. Feature Importance and Thermodynamic Attribution

TreeSHAP values computed across 5 random 1000-row seeds attribute 49.18% ± 0.12% of degradation importance to High-Pressure Compressor sensors: Ps30 (14.25% ± 0.23%), Nc (13.01% ± 0.24%), P30 (9.71% ± 0.12%), NRc (9.41% ± 0.16%), and T30 (2.78% ± 0.06%). Permutation importance yields a 62.22% HPC share.

Subassembly Station	Sensors Included	TreeSHAP Share %	Permutation Share %
High-Pressure Compressor (HPC)	Ps30, Nc, P30, NRc, T30	49.18% ± 0.12%	62.22%
Low-Pressure Turbine (LPT)	T50, W32, P15	17.86% ± 0.13%	13.67%
Combustion Chamber	phi, farB, htBleed	11.95% ± 0.08%	11.54%
Low-Pressure Compressor (LPC)	T24, BPR	11.56% ± 0.05%	7.08%
High-Pressure Turbine (HPT)	W31	5.36% ± 0.07%	2.74%
Fan & Inlet	T2, P2, Nf, NRf, dmd	4.09% ± 0.10%	2.75%
Engine-Level Pressure Ratio	epr	0.00% ± 0.00%	0.00%

Table VII. Subassembly Thermodynamic Feature Attribution Comparison.

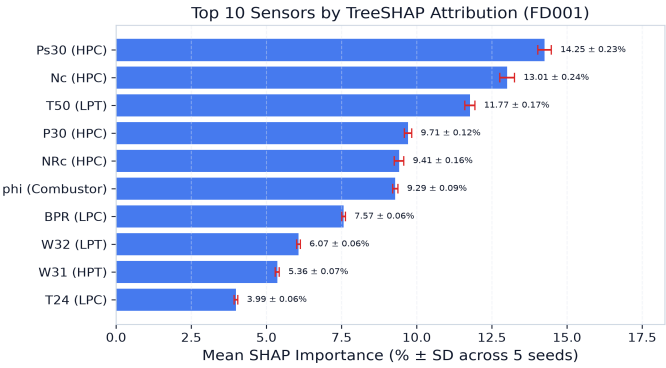


Fig. 6. Top 10 sensors ranked by TreeSHAP importance with standard deviation error bounds.

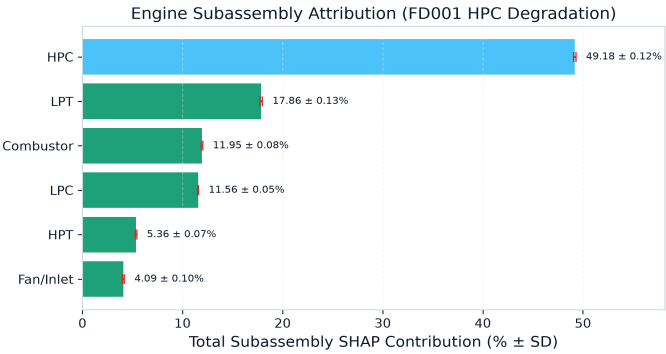


Fig. 7. Engine subassembly feature attribution under TreeSHAP and permutation cross-check.

VIII. DISCUSSION

The quantitative finding that HPC sensors account for 49.18% ± 0.12% of TreeSHAP attribution is consistent with the simulated physical fault mechanism of FD001, which models high-pressure compressor degradation. The higher HPC share observed in permutation importance (62.22%) reflects attribution sharing among collinear features; when correlated thermodynamic parameters (Ps30, P30, Nc) are perturbed independently, model performance degrades sharply.

The lead-time advantage of 21.0 median flight cycles observed for LSTM over Random Forest highlights the value of temporal context: multi-cycle sliding windows filter transient measurement noise that causes single-cycle snapshot classifiers to delay flagging degradation until sensor drift exceeds statistical baselines. However, we note that 82 of the 100 evaluated engines were part of the training gradient pool; on strictly held-out validation engines (83 to 100), the median advantage was 26.5 cycles, demonstrating consistent temporal sensitivity.

Isolation Forest achieved near-perfect critical recall (0.9970) but exhibited poor precision (0.1372), resulting from early false alarms (median trigger delay of -155.0 cycles). Consequently, unsupervised isolation trees are unsuitable as standalone early warnings but serve as secondary novelty detectors. Finally, the marked performance collapse on FD002 and FD004 (macro-F1 <= 0.1283) is hypothesized to arise from unnormalized operating conditions; without flight regime clustering, ambient pressure and Mach number variations dominate sensor telemetry.

From an operational perspective, combining snapshot classification with sequence regression enables tiered maintenance workflows. Immediate operational dispatch can query the lightweight Random Forest classifier for instantaneous health assurance, while depot-level overhaul planning leverages sequence-based LSTM regression to establish reliable shop-visit booking horizons.

IX. LIMITATIONS

1) *Synthetic Benchmark:* Evaluations are constrained to numerically simulated NPSS data; operational flight telemetry from airline carriers was not evaluated.
2) *Single-Regime Transfer Collapse:* Models trained on single-condition FD001 fail to generalize to multi-condition sets FD002 and FD004 due to the absence of operating condition normalization.
3) *Isolation Forest Precision:* Unsupervised Isolation Forest yields high false alarm rates on continuous non-stationary degradation curves.
4) *Computational Overhead:* Sequence windowing incurs higher memory footprint and inference latency compared to instantaneous single-cycle evaluation.

X. FUTURE WORK

Future research will investigate operating regime clustering (e.g., k-means flight condition partitioning) prior to sensor scaling to enable transferability across multi-regime datasets. In addition, physics-guided neural networks incorporating thermodynamic work balances and attention-based Transformer models will be explored for refined prognostic trajectory tracking.

REFERENCES

[1] A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage propagation modeling for aircraft engine run-to-failure simulation," in Proc. 2008 Int. Conf. Prognostics and Health Management (PHM08), Denver, CO, USA, 2008, pp. 1–9.

[2] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

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[3] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in Proc. 2008 Eighth IEEE Int. Conf. Data Mining (ICDM), Pisa, Italy, 2008, pp. 413–422.

[4] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[5] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems (NeurIPS 30), Long Beach, CA, USA, 2017, pp. 4765–4774.

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